Yashwant Malviya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.87
G S Jangdid Memorial School Jodhpur Raj	Class XII (Central Board of Secondary Education)	2024	80.60%
Central Academy School Pal Bye Pass Jodhpur Raj	Class X (Central Board of Secondary Education)	2022	90.20%

SKILLS

Programming & Scripting Languages: C++, Python, JavaScript, MySQL, HTML, CSS

Libraries: NumPy, Matplotlib, Pandas

Core Concepts: Object-Oriented Programming (OOP)

Tools: Git/GitHub, Solid Edge

SCHOLASTIC ACHIEVEMENTS

Data Structures & Algorithms: 800+ Problems Solved Across Coding Platforms, Including 700+ on LeetCode.

LeetCode: Peak Contest Rating of **1761**.

PROJECTS

Baat Cheet -- Real-Time Chat Application

Technologies Used: React, Node.js, Express.js, MongoDB, Socket.io, JWT

- Built an end-to-end real-time chat application with instant messaging capabilities.
- Developed scalable REST APIs using Express.js with JWT-based authentication and bcrypt password hashing.
- Implemented WebSocket-based real-time communication using Socket.io for instant message delivery and synchronization.
- Built a user presence system to track online/offline status and provide real-time typing indicators.
- Designed a responsive UI with custom CSS, ensuring a seamless experience across desktop and mobile devices.
- Leveraged MongoDB with Mongoose for persistent message storage and user management.

Algo Visualizer -- Interactive Algorithm Visualization Platform

Technologies Used: React, Vite, Node.js, Express, C++

- Built an end-to-end web application that visualizes **4 sorting algorithms** (Bubble Sort, Selection Sort, Insertion Sort, Merge Sort) and **5 graph algorithms** (BFS, DFS, Dijkstra's, Bellman-Ford, Floyd-Warshall) with real-time, step-by-step animations.
- Developed scalable REST APIs using **Express.js** and **C++ executables** for high-performance algorithm computation.
- Designed a React-based frontend with interactive controls and color-coded visual feedback to enhance the learning experience.

3D Train Model -- CAD Design & Additive Manufacturing

Technologies Used: Solid Edge, Creality 3D Printer

- Designed a detailed 3D CAD model of a train using **Solid Edge** with dimensionally accurate components and assembly features.
- Fabricated the physical prototype using a **Creality 3D printer** after preparing the model for additive manufacturing.
- Optimized the CAD model for printability by refining part geometry, tolerances, and support requirements.

Slayer Exciter -- Wireless Power Transmission Demonstrator

Technologies Used: Electronics, Circuit Design, TIP31C Transistor, Electromagnetic Induction

- Designed and built a Slayer Exciter circuit using a **TIP31C transistor**, primary and secondary copper coils, and discrete electronic components to generate high-frequency oscillations.
- Developed and tested a wireless power transmission prototype capable of illuminating **LEDs** and fluorescent bulbs without direct electrical connections through electromagnetic induction.
- Optimized coil winding, circuit connections, and component selection to achieve stable oscillations and reliable high-voltage operation.

RELEVANT COURSES

Python: Fundamentals of Scientific Computing, Computational Methods in Materials Science.

Mathematics: Differential Equation, Elementary Linear Algebra, Calculus I&II, Artifical Intelligence.

EXTRACURRICULAR

Qualified and Competed in State-Level: 10m Air Rifle & 10m Air Pistol Events.

Qualified for Round 2 of Flipkart GRID 8.0, a national-level software engineering competition, demonstrating strong problem-solving and coding skills.

POSITION OF RESPONSIBILITY

Manager & Assistant Manager, E-Cell: E-Cell empowers aspiring entrepreneurs by providing the resources, mentorship, and opportunities to turn ideas into ventures.